Cambridge International AS & A Level Physics

9702 Paper 4 – Synoptic Practice Question

The Sun: gravitation, kinetic theory and nuclear physics

Answer all questions. Show all working and give appropriate units. **40 marks**

Data

gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$	speed of light	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$	elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
atomic mass constant	$u = 1.66 \times 10^{-27}\,\text{kg}$	Coulomb constant	$k_\text{e} = 8.99 \times 10^{9}\,\text{N m}^2\,\text{C}^{-2}$
1 u	$= 9.32 \times 10^{2}\,\text{MeV c}^{-2}$	1 year	$= 3.16 \times 10^{7}\,\text{s}$

The following measured or modelled solar data may be used.

mean Earth–Sun distance, r	$1.50 \times 10^{11}\,\text{m}$	solar radius, $R_\odot$	$6.96 \times 10^{8}\,\text{m}$
Earth orbital period, T	365.256 days	present luminosity, $L_\odot$	$3.85 \times 10^{26}\,\text{W}$
central temperature	$1.57 \times 10^{7}\,\text{K}$	central density	$1.62 \times 10^{5}\,\text{kg m}^{-3}$
present age	4.57×10^{9} years	initial H mass fraction, X	0.71

1 The mass of the Sun

Treat the Earth's orbit as circular and neglect the mass of the Earth compared with the mass $M_\odot$ of the Sun.

(a) By equating gravitational force to centripetal force, show that

$$M_\odot = \frac{4\pi^2 r^3}{GT^2}.$$

[2]

(b) Use the data to calculate $M_\odot$. **[3]**

(c) State why neglecting the mass of the Earth has a negligible effect on the answer. **[1]**

2 A kinetic-theory model of the solar core

The solar core is modelled as a fully ionised ideal gas. For the mixture of ions and electrons, the mean mass per free particle is $0.61u$.

(a) Calculate the mean translational kinetic energy of a particle at the central temperature. Give your answer in J and in keV. **[2]**

(b) Calculate the root-mean-square speed of a proton at this temperature. **[2]**

(c) Use the kinetic-theory equation $p = nkT$ to estimate the gas pressure at the centre of the Sun. **[2]**

(d) Two protons must approach to a separation of about 1.00×10^{-15} m before the strong nuclear force is significant.

Calculate their electric potential energy at this separation. Hence explain why fusion can occur in the Sun even though the mean kinetic energy calculated in (a) is much smaller than this potential energy. **[3]**

(e) State one limitation of treating the solar core as an ideal gas. **[1]**

3 Nuclear energy in the Sun

The main source of solar energy is the proton–proton chain. Three stages are represented below. Copy and complete the equations.

(a)

$$
\begin{aligned}
{}^{1}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} &\longrightarrow {}^{2}_{1}\mathrm{H} + \ \cdots\ + \nu_e, \\
{}^{2}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} &\longrightarrow {}^{3}_{2}\mathrm{He} + \ \cdots\ , \\
{}^{3}_{2}\mathrm{He} + {}^{3}_{2}\mathrm{He} &\longrightarrow {}^{4}_{2}\mathrm{He} + \ \cdots\ .
\end{aligned}
$$

[3]

(b) The relevant atomic masses are

$$m({}^{1}\mathrm{H}) = 1.007825u, \qquad m({}^{4}\mathrm{He}) = 4.002603u.$$

The overall result of the chain is the conversion of four hydrogen atoms into one helium atom, with neutrinos carrying away a small part of the released energy.

Calculate the energy released per completed chain, first in MeV and then in J. **[3]**

(c) Assume initially that all the energy calculated in (b) contributes to the solar luminosity. Calculate the number of completed chains per second in the Sun. **[2]**

(d) Calculate the mass of hydrogen consumed per second. **[2]**

(e) Calculate the mass converted to energy per second at the present luminosity. **[1]**

4 Evolution of the Sun

Fig. 1 shows results from a standard solar model. The luminosity $L(t)$ is divided by the measured present luminosity $L_\odot$. The curve uses the published model data listed below the graph.

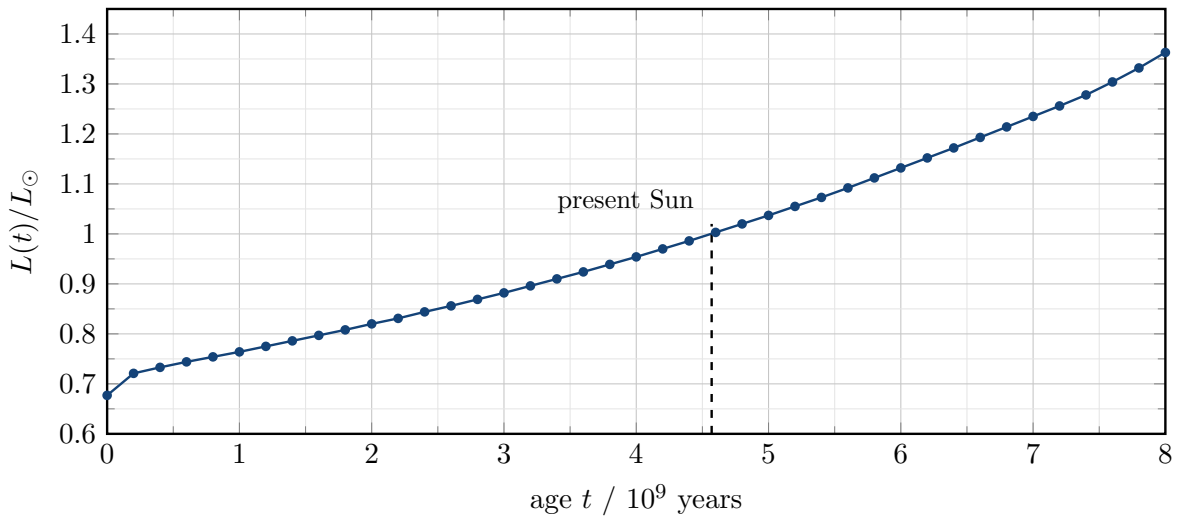


Fig. 1

Selected plotted values:

$(t/\mathrm{Gyr}, L/L_\odot) = (0, 0.677), (1, 0.764), (2, 0.820), (3, 0.882), (4, 0.954), (4.6, 1.003), (5, 1.037), (6, 1.132), (7, 1.235), (8, 1.363).$

(a) Use Fig. 1 to determine $L/L_\odot$ at ages 2.00×10^9 years and 6.00×10^9 years. **[2]**

(b) The area under the graph from $t = 0$ to the present age is approximately 3.88×10^9 years.

Use this result to estimate the total energy radiated by the Sun during its main-sequence life so far. **[3]**

(c) Calculate the mass equivalent of this radiated energy. **[2]**

5 The lifetime of the Sun

In a simplified model, 13% of the solar mass is at sufficiently high temperature for hydrogen fusion. Of this mass, the initial hydrogen fraction is 0.71. Fusion converts 0.70% of the mass of hydrogen into energy.

(a) Assume that the luminosity remains equal to its present value. Estimate the total time for which the Sun could remain on the main sequence. **[4]**

(b) Hence estimate the remaining main-sequence lifetime of the Sun. **[1]**

(c) Use Fig. 1 to explain why the calculation in (a) is only an approximate estimate. **[1]**

[Total: 40]

Guided solutions and marking guidance

The mark allocations indicate suitable marking points; alternative correct methods should receive credit.

1 The mass of the Sun

(a) For the Earth, gravitational force supplies centripetal force:

$$\frac{GM_\odot m}{r^2} = \frac{mv^2}{r}, \qquad v = \frac{2\pi r}{T}.$$

Cancelling m and substituting for v gives

$$M_\odot = \frac{4\pi^2 r^3}{GT^2}.$$

[1 for force equation; 1 for substitution and result]

(b) Convert the period to seconds:

$$T = 365.256 \times 86400 = 3.16 \times 10^7\,\text{s}.$$

$$M_\odot = \frac{4\pi^2(1.496\times10^{11})^3}{(6.6743\times10^{-11})(3.1558\times10^7)^2} = \boxed{1.99\times10^{30}\,\text{kg}}.$$

[conversion; substitution; answer]

(c) The Earth is about 3×10^{-6} of the Sun's mass, so including it changes the result by only a few parts per million. **[1]**

2 A kinetic-theory model of the solar core

(a)

$$\langle E_k\rangle = \frac{3}{2}kT = \frac{3}{2}(1.380649\times10^{-23})(1.57\times10^7) = \boxed{3.25\times10^{-16}\,\text{J}}.$$

Since $1.00\,\text{eV} = 1.60\times10^{-19}\,\text{J}$,

$$\langle E_k\rangle = 2.03\times10^3\ \text{eV} = \boxed{2.03\,\text{keV}}.$$

(b)

$$\frac{1}{2}m_p c_{\text{rms}}^2 = \frac{3}{2}kT, \qquad c_{\text{rms}} = \sqrt{\frac{3kT}{m_p}} = \boxed{6.24\times10^5\,\text{m s}^{-1}}.$$

(c) The number density of free particles is

$$n = \frac{\rho}{0.61u} = \frac{1.62\times10^5}{0.61(1.66054\times10^{-27})} = 1.60\times10^{32}\ \text{m}^{-3}.$$

Hence

$$p = nkT = \boxed{3.47\times10^{16}\,\text{Pa}}.$$

(d)

$$E_{\text{electric}} = \frac{k_e e^2}{r} = \frac{(8.988\times10^9)(1.60218\times10^{-19})^2}{1.0\times10^{-15}} = \boxed{2.31\times10^{-13}\,\text{J}} = \boxed{1.44\,\text{MeV}}.$$

This is about 700 times the mean kinetic energy. A small fraction of particles has energy well above the mean because speeds have a Maxwell–Boltzmann distribution. More importantly, quantum tunnelling gives protons a finite probability of passing through the Coulomb barrier without having the full classical barrier energy. **[calculation; comparison; explanation]**

(e) For example: Coulomb interactions mean the particles are not perfectly non-interacting; radiation pressure also contributes; or density and temperature vary throughout the core. **[1]**

3 Nuclear energy in the Sun

(a)

$$
\begin{aligned}
{}^{1}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} &\longrightarrow {}^{2}_{1}\mathrm{H} + {}^{0}_{+1}\mathrm{e} + \nu_e, \\
{}^{2}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} &\longrightarrow {}^{3}_{2}\mathrm{He} + \gamma, \\
{}^{3}_{2}\mathrm{He} + {}^{3}_{2}\mathrm{He} &\longrightarrow {}^{4}_{2}\mathrm{He} + 2{}^{1}_{1}\mathrm{H}.
\end{aligned}
$$

[one mark for each missing product]

(b)

$$\Delta m = 4(1.007825) - 4.002603 = 0.028697u.$$

$$Q = 0.028697 \times 931.5 = \boxed{2.67 \times 10^{1}\,\mathrm{MeV}}.$$

$$Q = 26.7 \times 10^{6}(1.60218 \times 10^{-19}) = \boxed{4.28 \times 10^{-12}\,\mathrm{J}}.$$

(c)

$$\text{chains per second} = \frac{L_\odot}{Q} = \frac{3.846 \times 10^{26}}{4.28 \times 10^{-12}} = \boxed{8.99 \times 10^{37}\,\mathrm{s^{-1}}}.$$

(d) Four hydrogen atoms are used per completed chain:

$$\dot{m}_{\mathrm{H}} = 4(1.007825u)(8.99 \times 10^{37}) = \boxed{6.02 \times 10^{11}\,\mathrm{kg\,s^{-1}}}.$$

This is about 600 million tonnes per second.

(e)

$$\dot{m} = \frac{L_\odot}{c^2} = \frac{3.846 \times 10^{26}}{(2.998 \times 10^{8})^2} = \boxed{4.28 \times 10^{9}\,\mathrm{kg\,s^{-1}}}.$$

4 Evolution of the Sun

(a) From the graph,

$$\boxed{L/L_\odot \approx 0.82 \text{ at } 2.00 \times 10^9 \text{ years}}, \qquad \boxed{L/L_\odot \approx 1.13 \text{ at } 6.00 \times 10^9 \text{ years}}.$$

(b) The graph area has units of time because the vertical axis is dimensionless. Thus

$$\begin{aligned} E &= L_\odot \times (3.88 \times 10^9 \text{ years}) \\ &= (3.846 \times 10^{26})(3.88 \times 10^9)(3.156 \times 10^7) \\ &= \boxed{4.71 \times 10^{43} \text{ J}}. \end{aligned}$$

(c)

$$m = \frac{E}{c^2} = \frac{4.71 \times 10^{43}}{(2.998 \times 10^8)^2} = \boxed{5.24 \times 10^{26} \text{ kg}}.$$

This is about 2.6×10^{-4} of the present solar mass.

5 The lifetime of the Sun

(a) The mass of hydrogen available for fusion is

$$m_\text{H} = 0.13(0.71)M_\odot = 0.13(0.71)(1.99 \times 10^{30}) = 1.84 \times 10^{29} \text{ kg}.$$

The energy available is

$$E = 0.0070 m_\text{H} c^2 = 1.16 \times 10^{44} \text{ J}.$$

Thus

$$t = \frac{E}{L_\odot} = 3.01 \times 10^{17} \text{ s} = \boxed{9.50 \times 10^9 \text{ years}}.$$

(b)

$$t_\text{remaining} = 9.5 - 4.57 = \boxed{4.90 \times 10^9 \text{ years}}.$$

(c) Fig. 1 shows that luminosity increases with age, so it has not remained at its present value. The estimated lifetime also depends on the simplified fraction of the Sun that can fuse hydrogen and neglects energy carried away by neutrinos. **[any one relevant point]**

Data provenance. Solar mass, radius, luminosity and central model values are consistent with the NASA/GSFC Sun Fact Sheet. The age and description of the fusion core are consistent with NASA's "Our Sun: Facts". The luminosity–age and radius–age points are from J. N. Bahcall, M. H. Pinsonneault and S. Basu, *The Astrophysical Journal* **555** (2001), Tables 1 and 2. Fundamental constants use CODATA values. Values have been rounded to an appropriate precision for an A-level exercise.

Variant 2 – Higher Difficulty

The internal structure, radiation and evolution of the Sun

Answer all questions. This variant contains no quantum-tunnelling questions. **45 marks**

Additional data for Variant 2

proton mass	$m_p = 1.67 \times 10^{-27}\,\text{kg}$	electron mass	$m_e = 9.11 \times 10^{-31}\,\text{kg}$
Stefan constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$	radiation constant	$a = 7.57 \times 10^{-16}\,\text{J m}^{-3}\,\text{K}^{-4}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$	solar irradiance at Earth	$S = 1.36 \times 10^{3}\,\text{W m}^{-2}$

Use $M_\odot = 1.99 \times 10^{30}\,\text{kg}$, $R_\odot = 6.96 \times 10^{8}\,\text{m}$ and the constants given at the start of Variant 1.

6 Gravitation and a model of the solar interior

(a) Calculate the mean density $\bar{\rho}$ of the Sun. **[2]**

(b) Calculate the gravitational field strength g_s at the solar surface. **[2]**

(c) A simple model treats the Sun as a sphere of uniform density. It estimates the central pressure using

$$P_c \approx \frac{1}{2}\bar{\rho} g_s R_\odot .$$

Show that this may be written as

$$P_c \approx \frac{3GM_\odot^2}{8\pi R_\odot^4},$$

and calculate P_c. **[3]**

(d) The central pressure predicted by a detailed solar model is approximately $2.50 \times 10^{16}\,\text{Pa}$. Suggest why this is much larger than the result from the uniform-density model. **[2]**

(e) In an orbital determination, the fractional uncertainties in r and T are 0.0020% and 0.00010%, respectively. Use

$$M_\odot = \frac{4\pi^2 r^3}{GT^2}$$

to estimate the percentage uncertainty in $M_\odot$. Treat G as exact. **[1]**

7 Kinetic theory and radiation pressure

Continue to use the uniform-density pressure calculated in 6(c). The mean mass per free particle is $0.61u$.

(a) Use $P = \rho kT/(0.61u)$ to calculate the temperature predicted by this simple model. **[3]**

(b) Calculate the root-mean-square speed of a proton at this temperature. Without further calculation, determine the ratio $c_{\text{rms,electron}}/c_{\text{rms,proton}}$. **[2]**

(c) The pressure of thermal radiation is

$$P_{\text{rad}} = \frac{1}{3} a T^4.$$

Calculate P_{rad} and determine P_{rad}/P_c. Comment on the importance of radiation pressure in this model. **[3]**

8 Solar irradiance, fusion and neutrinos

The solar irradiance S is the power received per unit area on a surface normal to the Sun's rays at the mean Earth–Sun distance.

(a) Show that the solar luminosity is given by $L = 4\pi r^2 S$ and calculate L. **[2]**

(b) The energy release calculated from the atomic mass difference for one completed proton–proton chain is 2.67×10^1 MeV. Assume that neutrinos carry away 2.0% of this energy.

Calculate the energy per chain that is available to produce the measured luminosity, in J. **[3]**

(c) Calculate the number of completed proton–proton chains per second. **[2]**

(d) Two electron-neutrinos are produced in each completed chain. Assuming isotropic emission, calculate the neutrino flux at the Earth in $\mathrm{m^{-2}\,s^{-1}}$. **[2]**

(e) A simplified water detector contains 2.00×10^7 kg of water. Each water molecule contains 10 electrons. For the neutrinos considered, take the effective interaction cross-section per electron to be $9.00 \times 10^{-49}\,\mathrm{m^2}$.

Estimate the number of neutrino interactions per second in the detector. **[3]**

9 Radius, luminosity and effective temperature

Fig. 2 contains published standard-solar-model values of radius against age. Radius is normalised to its present value. The luminosity values required in this question may be read from Fig. 1.

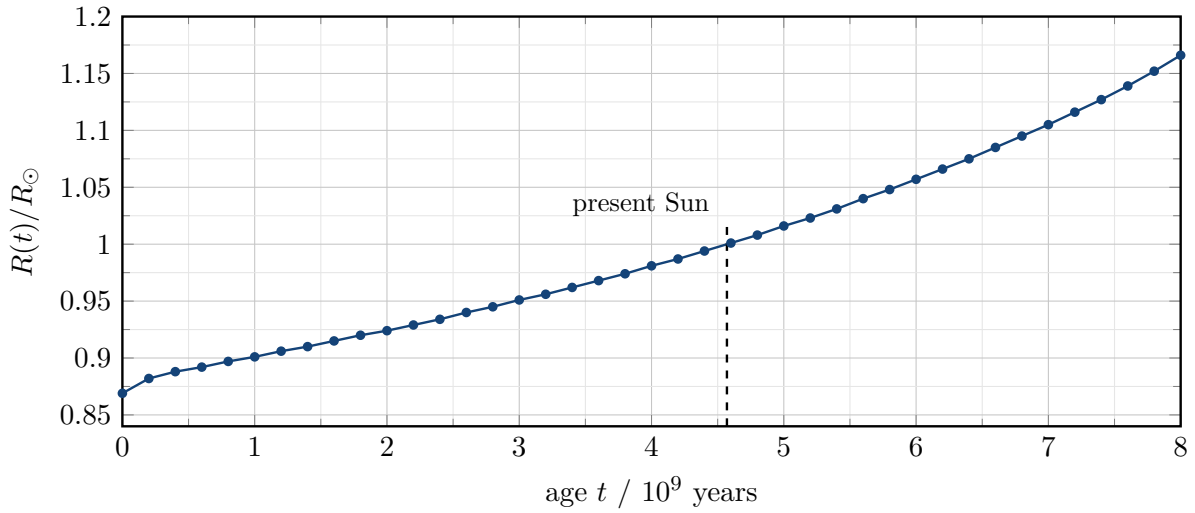


Fig. 2

(a) Determine $R/R_\odot$ at ages 2.00×10^9 years and 7.00×10^9 years. **[2]**

(b) The Stefan–Boltzmann law is $L = 4\pi R^2 \sigma T_{\mathrm{eff}}^4$. Show that

$$\frac{T_{\mathrm{eff}}(t)}{T_{\mathrm{eff},0}} = \left[\frac{L(t)/L_\odot}{\{R(t)/R_\odot\}^2}\right]^{1/4}.$$

Use Figs. 1 and 2 to determine this temperature ratio at ages zero and 8.00×10^9 years. **[4]**

(c) The effective temperature changes very little between ages zero and 8.00×10^9 years. Explain how the luminosity can nevertheless increase substantially. **[2]**

10 A refined lifetime estimate

The total fusion energy available in the simplified fuel model of Question 5 is equal to the energy emitted by the present Sun in 9.52×10^9 years. A linear approximation to Fig. 1 is

$$\frac{L(t)}{L_\odot} = 0.677 + 0.071t,$$

where t is measured in billions of years.

(a) Show that the energy emitted by age t, expressed in units of "present solar luminosity $\times 10^9$ years", is

$$0.677t + 0.0355t^2.$$

[2]

(b) Hence solve

$$0.677t + 0.0355t^2 = 9.52$$

to estimate the total main-sequence lifetime. **[3]**

(c) Estimate the remaining main-sequence lifetime. **[1]**

(d) Use Fig. 1 to state whether the linear approximation is likely to overestimate or underestimate the lifetime. Explain your answer. **[1]**

[Variant 2 total: 45]

Variant 2 – Guided solutions

6 Gravitation and a model of the solar interior

(a)
$$\bar{\rho} = \frac{M_\odot}{\frac{4}{3}\pi R_\odot^3} = \boxed{1.41 \times 10^3\,\mathrm{kg\,m^{-3}}}.$$

(b)
$$g_s = \frac{GM_\odot}{R_\odot^2} = \boxed{2.74 \times 10^2\,\mathrm{N\,kg^{-1}}}.$$

(c) Substituting $\bar{\rho} = 3M_\odot/(4\pi R_\odot^3)$ and $g_s = GM_\odot/R_\odot^2$ gives
$$P_c \approx \frac{1}{2}\left(\frac{3M_\odot}{4\pi R_\odot^3}\right)\left(\frac{GM_\odot}{R_\odot^2}\right)R_\odot = \frac{3GM_\odot^2}{8\pi R_\odot^4}.$$
$$P_c = \boxed{1.35 \times 10^{14}\,\mathrm{Pa}}.$$

(d) The real Sun is strongly centrally concentrated rather than uniform. Much of the overlying mass compresses a small, dense core, and both density and the gravitational field vary with radius. **[any two relevant points]**

(e)
$$\frac{\Delta M}{M} \approx 3\frac{\Delta r}{r} + 2\frac{\Delta T}{T} = 3(0.0020\%) + 2(0.00010\%) = \boxed{0.0062\%}.$$

7 Kinetic theory and radiation pressure

(a)
$$T = \frac{P_c(0.61u)}{\bar{\rho}k} = \boxed{7.00 \times 10^6\,\mathrm{K}}.$$

The value is lower than the detailed-model central temperature because the uniform-density model greatly underestimates the central pressure.

(b)
$$c_{\mathrm{rms,p}} = \sqrt{\frac{3kT}{m_p}} = \boxed{4.16 \times 10^5\,\mathrm{m\,s^{-1}}}.$$

At the same temperature $c_{\mathrm{rms}} \propto m^{-1/2}$, so
$$\frac{c_{\mathrm{rms,e}}}{c_{\mathrm{rms,p}}} = \sqrt{\frac{m_p}{m_e}} = \boxed{42.9}.$$

(c)
$$P_{\mathrm{rad}} = \frac{1}{3}(7.566 \times 10^{-16})(7.00 \times 10^6)^4 = \boxed{6.06 \times 10^{11}\,\mathrm{Pa}}.$$
$$\frac{P_{\mathrm{rad}}}{P_c} = \boxed{4.5 \times 10^{-3}}.$$

Radiation pressure is less than 1% of the estimated gas pressure, so it is a small correction in this model. **[1]**

8 Solar irradiance, fusion and neutrinos

(a) The power crosses a sphere of area $4\pi r^2$, so
$$L = 4\pi r^2 S = 4\pi(1.496 \times 10^{11})^2(1361) = \boxed{3.83 \times 10^{26}\,\mathrm{W}}.$$

(b)

$$Q_{\text{available}} = 0.980(26.73) = 26.20 \text{ MeV},$$

$$Q_{\text{available}} = 26.20 \times 10^6 (1.60218 \times 10^{-19}) = \boxed{4.20 \times 10^{-12} \text{ J}}.$$

(c)

$$N = \frac{L}{Q_{\text{available}}} = \boxed{9.12 \times 10^{37} \text{ s}^{-1}}.$$

(d)

$$\Phi_\nu = \frac{2N}{4\pi r^2} = \boxed{6.49 \times 10^{14} \text{ m}^{-2} \text{ s}^{-1}}.$$

(e) The number of target electrons is

$$N_e = \frac{2.0 \times 10^7}{0.0180} N_{\text{A}}(10) = 6.69 \times 10^{33}.$$

The interaction rate is

$$R = \Phi_\nu N_e \sigma_\nu = (6.49 \times 10^{14})(6.69 \times 10^{33})(9.0 \times 10^{-49}) = \boxed{3.90 \text{ s}^{-1}}.$$

9 Radius, luminosity and effective temperature

(a)

$$\boxed{R/R_\odot = 0.924 \text{ at } 2.00 \times 10^9 \text{ years}}, \qquad \boxed{R/R_\odot = 1.105 \text{ at } 7.00 \times 10^9 \text{ years}}.$$

(b) Dividing $L(t) = 4\pi R(t)^2 \sigma T(t)^4$ by the present-day equation gives

$$\frac{L(t)}{L_\odot} = \left(\frac{R(t)}{R_\odot}\right)^2 \left(\frac{T(t)}{T_{\text{eff},0}}\right)^4.$$

Rearrangement gives the stated expression. At zero age,

$$\frac{T}{T_{\text{eff},0}} = \left(\frac{0.677}{0.869^2}\right)^{1/4} = \boxed{0.973}.$$

At 8.00×10^9 years,

$$\frac{T}{T_{\text{eff},0}} = \left(\frac{1.363}{1.166^2}\right)^{1/4} = \boxed{1.001 \approx 1.00}.$$

(c) The radiating surface area is proportional to R^2. The radius increases considerably, so a nearly unchanged surface temperature can still produce a substantially larger luminosity. **[2]**

10 A refined lifetime estimate

(a) For a linear change, the mean normalised luminosity is the average of the initial and final values:

$$\frac{\overline{L}}{L_\odot} = \frac{0.677 + (0.677 + 0.071t)}{2} = 0.677 + 0.0355t.$$

Multiplication by t gives $0.677t + 0.0355t^2$. **[2]**

(b) Solving

$$0.0355t^2 + 0.677t - 9.52 = 0$$

gives

$$t = \frac{-0.677 + \sqrt{0.677^2 + 4(0.0355)(9.52)}}{2(0.0355)} = \boxed{9.41 \times 10^9 \text{ years}}.$$

The negative root has no physical meaning.

(c)

$$t_{\text{remaining}} = 9.41 - 4.57 = \boxed{4.84 \times 10^9 \text{ years}}.$$

(d) Beyond about 5.00×10^9 years, Fig. 1 curves above the straight-line model, so the model underestimates the future luminosity. Fuel would therefore be used more rapidly than predicted, and the linear model **overestimates** the lifetime. **[1]**